

第 1 章 Maximum Likelihood Estimation

1.1 Likelihood function and the ML principle

Let $f(y; \theta_0)$ denote the probability density function of the random variable y given θ_0 . Our objective is to estimate the true parameter vector θ_0 . The joint density of n iid observations of y is

$$f(y_1, \dots, y_n | \theta_0) = \prod_{i=1}^n f(y_i; \theta_0) \quad (1)$$

Now taking $f(y_i; \theta)$ as a function of θ given $y_1, \dots, y_n$ and write

$$L(\theta | y_1, \dots, y_n) = \prod_{i=1}^n f(y_i; \theta) \quad (2)$$

gives the **likelihood function**, the likelihood that the population parameter is θ given the observed sample.

1.1.1 ML principle

The **Maximum Likelihood (ML) principle** suggests that estimators of the unknown parameters are obtained by **maximizing $L(\theta)$ with respect to θ** . If the random variable y is discrete, $\hat{\theta}$ yields the value of θ that maximizes the probability of observing the sample that actually occurred.

Usually, $\hat{\theta}$ can be obtained by solving the likelihood equation

$$\frac{\partial \log L(\theta)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \quad (3)$$

where $\log L(\theta | y_1, \dots, y_n) = \sum_{i=1}^n \log f(y_i; \theta)$. Occasionally the ML estimator is not unique and $\log L(\theta)$ may have only one global maximum, but multiple local maxima.

The **regularity conditions** are the following:

1. The first three derivatives of $\log f(y|\theta)$ with respect to θ are continuous and finite for almost all y and for all θ .
2. For all values of θ , $|\frac{\partial^3 \log f(y|\theta)}{\partial \theta_j \partial \theta_k \partial \theta_l}|$ is limited by a function that has finite expectations.
3. The domain of y does not depend on θ . **For example of the uniform distribution, since the density function depends on θ_0 , the theorem does not hold.**
4. θ is an interior point to the compact parameter space Θ .

1.1.2 Bartlett Identities

Define the **score vector** $S(\theta)$ and the **Hessian matrix** $H(\theta)$ as

$$S(\theta) = \frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial \log f(y_i|\theta)}{\partial \theta} = \sum_{i=1}^n s_i(\theta) \quad (4)$$

$$H(\theta) = \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^n \frac{\partial^2 \log f(y_i|\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^n H_i(\theta) \quad (5)$$

- The **First Bartlett identity** states that $E[s_i(\theta_0)] = 0$. Hence, $E[S(\theta_0)] = 0$.
- The **Second Bartlett identity** states that $\text{Var}[s_i(\theta_0)] = -E[H_i(\theta_0)]$.

证明.

$$\text{Var} \left[\frac{\partial \log f(y_i|\theta)}{\partial \theta} \right] = -E \left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'} \right]$$

where y_i is a random variable. □

$\text{Var}[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)']$ defines the **Fisher's information matrix** denoted as $\mathcal{I}(\theta_0)$. Hence, the result $\text{Var}[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)'] = -E[H_i(\theta_0)]$ is also called the **information matrix identity**.

证明.

$$\begin{aligned} \text{Var}[s_i(\theta)] &= E \left[\left[s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0, \text{ First Bartlett}} \right] \times \left[s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0, \text{ First Bartlett}} \right]' \right] \\ &= E[s_i(\theta_0)s_i(\theta_0)'] \end{aligned}$$
□

1.2 Properties of MLE

Under the assumed regularity conditions of MLE possesses the following properties:

1. **Consistency** meaning $\text{plim} \hat{\theta} = \theta_0$
2. **Asymptotic normality** meaning $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$. The goal is to estimate the $[\mathcal{I}(\theta_0)]^{-1}$ matrix (Fisher's information matrix) $\rightarrow$ need to go to a known distribution (normal).
3. **Asymptotic efficiency** meaning that if $\tilde{\theta}$ is a regular consistent asymptotically normal estimator such that $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$, then $\Omega - [\mathcal{I}(\theta_0)]^{-1}$ is positive semi-definite, i.e., under these RC, the MLE asymptotically achieves the **Cramer-Rao** lower bound which is given by $[\mathcal{I}(\theta_0)]^{-1}$. Information matrix is smaller than Ω in the matrix sense, and the information matrix is the lower bound.
4. **Invariance** meaning that if $c(\theta)$ is a continuous and continuously differentiable one-to-one function, the MLE of $\gamma = c(\theta_0)$ is $c(\hat{\theta})$.

Example

In the Bernoulli case $\mathcal{I}(\theta_0)^{-1} = \theta_0(1 - \theta_0)$, therefore, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \theta_0(1 - \theta_0))$.

证明. We begin by writing the log-likelihood function,

$$\log f(y_i|\theta) = y_i \log(\theta) + (1 - y_i) \log(1 - \theta)$$

Now finding the score vector,

$$\begin{aligned} s_i(\theta) &= \frac{\partial \log f(y_i|\theta)}{\partial \theta} \\ &= \frac{y_i}{\theta} - \frac{(1 - y_i)}{(1 - \theta)} \\ &= \frac{y_i}{\theta} - \frac{1}{1 - \theta} + \frac{y_i}{1 - \theta} \\ &= y_i \left(\frac{1}{\theta} + \frac{1}{1 - \theta} \right) - \frac{1}{\theta} \\ &= y_i \left(\frac{1 - \theta + \theta}{\theta(1 - \theta)} \right) - \frac{1}{1 - \theta} \\ &= y_i \frac{1}{\theta(1 - \theta)} - \frac{1}{1 - \theta} \\ &= \frac{y_i}{\theta(1 - \theta)} - \frac{1}{1 - \theta} \end{aligned}$$

Now finding the Fisher's information matrix,

$$\text{Var}(s_i(\theta_0)) = \text{Var} \left[\frac{y_i}{\theta_0(1 - \theta_0)} - \underbrace{\frac{1}{1 - \theta_0}}_{=0, \text{ constant}} \right] = \text{Var} \left[\underbrace{\frac{y_i}{\theta_0(1 - \theta_0)}}_{\text{constant}} \right] = \frac{1}{\theta_0^2(1 - \theta_0)^2} \underbrace{\text{Var}[y_i]}_{(1)}$$

(1):

$$\text{Var}[y_i] = \underbrace{E[y_i^2]}_{(2)} - \underbrace{E[y_i]^2}_{(3)} = \theta_0 - \theta_0^2 = \theta_0(1 - \theta_0)$$

(2):

$$E[y_i^2] = 1^2 \underbrace{P(y_i = 1|\theta_0)}_{=\theta_0} + 0^2 \underbrace{p(y_i = 0|\theta_0)}_{(1-\theta_0)} = \theta_0$$

(3):

$$E[y_i] = 1 \underbrace{P(y_i = 1|\theta_0)}_{=\theta_0} + 0 \underbrace{p(y_i = 0|\theta_0)}_{(1-\theta_0)} = \theta_0$$

$$\text{Var}(s_i(\theta_0)) = \frac{1}{\theta_0^2(1 - \theta_0)^2} \theta_0(1 - \theta_0) = \frac{1}{\theta_0(1 - \theta_0)} = \mathcal{I}(\theta_0)$$

Now finding the inverse of the information matrix,

$$\mathcal{I}(\theta_0)^{-1} = \theta_0(1 - \theta_0)$$

We now conclude,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \theta_0(1 - \theta_0))$$

□

Now, don't forget that θ_0 is not known, so we want to estimate the information matrix.

1.2.1 Estimators of the information matrix

There are three commonly used estimators of $\mathcal{I}(\theta_0)$:

- **Expected Information:** If the form of the expected values of the second derivatives of the log-likelihood function is known, then we can evaluate $\mathcal{I}(\theta)$ at $\hat{\theta}$. [This rarely happens in practice.](#)
- **Observed information:** Simply use $-\hat{H}(\hat{\theta})$.

证明.

$$\begin{aligned} \mathcal{I}(\theta_0) &= \text{Var}[s_i(\theta_0)] = -E[H(\theta_0)] \\ &= \text{Var} \left[\frac{\partial \log f(y_i|\theta_0)}{\partial \theta_0} \right] \\ &= -E \left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'} \right] \\ &= \frac{1}{\text{avg. } N} \sum_{i=1}^n \left(\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'} \right) \\ &= -\hat{H}(\hat{\theta}) \end{aligned}$$

□

- **Outer Product of the Gradient (OPG):** Because of the information matrix identity, we can also use $n^{-1} \sum_{i=1}^n s_i(\hat{\theta}) s_i(\hat{\theta})'$. [The OPG is notorious for its poor performance with finite samples.](#)

证明.

$$\begin{aligned} \text{Var}[s_i(\theta_0)] &= E[s_i(\theta_0) s_i(\theta_0)'] \\ &= \frac{1}{N} \sum_{i=1}^N s_i(\theta_0) s_i(\theta_0)' \\ &= \frac{1}{N} \sum_{i=1}^N s_i(\hat{\theta}) s_i(\hat{\theta})' \end{aligned}$$

□

1.3 Regressors

Suppose the joint distribution of y and x depends on α_0 , giving

$$f(y, x|\alpha_0) = f(y|x, \alpha_0)g(x|\alpha_0) \quad (6)$$

$$f(y|x, \alpha_0) = \frac{f(y, x, \alpha_0)}{g(x, \alpha_0)} \quad (7)$$

$$\underbrace{f(y, x, \alpha_0)}_{\text{joint}} = \underbrace{f(y|x, \alpha_0)}_{\text{conditional}} \underbrace{g(x, \alpha_0)}_{\text{marginal}} \quad (8)$$

and suppose α_0 can be divided into θ_0 and δ_0 , so that with exogeneity of x , $f(y, x|\alpha_0) = f(y|x_i, \theta_0)g(x_i|\delta_0)$.

For an iid sample $\{(y_i, x_i)\}_{i=1}^n$,

$$\log L(\theta, \delta|y_i, x_i) = \sum_{i=1}^n \log f(y_i, x_i|\alpha_0) = \sum_{i=1}^n \log f(y_i|x_i, \theta) + \sum_{i=1}^n \log g(x_i|\delta) \quad (9)$$

We are under the assumption that we are only interested in θ_0 . Now this assumption that we don't care about δ makes our lives easier because

$$\frac{\partial L(\theta, \delta|y_i, x_i)}{\partial \theta} \Big|_{\hat{\theta}} = 0 \Leftrightarrow \frac{\partial \sum_{i=1}^n \log f(y_i|x_i, \theta_i)}{\partial \theta} \Big|_{\hat{\theta}} = 0$$

The part of $g(x, \delta)$ can be ignored now, only focus on the conditional density function. However, now $E[-H_i(\theta_0)] = \mathcal{I}(\theta_0)$.

1.4 Interpretation and inference in misspecified models

If the likelihood function is **misspecified**, the MLE is generally **inconsistent** for the parameters of interest. However, under the general conditions, $\text{plim} \hat{\theta} = \theta^*$, where the **pseudo-true value** θ^* minimizes the Kullback-Leibler divergence where $f_0(y)$ is the true distribution of data.

定义 1.4.1. Kullback-Leibler divergence, also called *relative entropy*, is a measure of how one probability distribution is different from a second, reference probability distribution.

This means that the MLE leads to the best approximation in the Kullback-Leibner sense to the true density. However, because Fisher's information matrix identity does not hold, the asymptotic covariance matrix is given by:

$$A^{-1}BA^{-1} \quad A = E[-H_i(\theta^*)] \quad B = E[s_i(\theta^*)s_i(\theta^*)'] \quad (10)$$

$f(y|\theta^*)$ is the best approximation given the wrong space but is still wrong.

1.5 Hypothesis testing

Considering a general set of restrictions to be tested

$$H_0 : h(\theta_0) = 0 \quad (11)$$

where

- θ : vector of parameters in the model and the true value is given by θ_0 .
- $h(\theta)$ is a $d \times 1$ vector of restrictions.
- $\hat{\theta}$ is the unrestricted MLE where its value maximizes $L(\theta)$.
- $\tilde{\theta}$ is the restricted MLE where its value maximizes $L(\theta)$ and satisfies $h(\theta) = 0$.

1.5.1 The Wald Test

Compares $h(\hat{\theta})$ with 0 since $h(\tilde{\theta}) = 0$. The test statistic is

$$\mathcal{W} = n \times h(\hat{\theta})' \left[G(\hat{\theta})' [\mathcal{I}(\hat{\theta})]^{-1} G(\hat{\theta}) \right]^{-1} h(\hat{\theta}) \quad (12)$$

where $G(\theta) = \frac{\partial h(\theta)}{\partial \theta}$.

Under the null hypothesis,

$$\mathcal{W} \xrightarrow{d} \chi^2(d) \quad (13)$$

Wald test is not invariant to how restrictions are formulated. Equivalent non-linear ($\beta_0/(1-\alpha_0) = 1$) and linear ($\beta_0 + \alpha_0 - 1 = 0$) restriction may lead to different values of $\mathcal{W}$.

1.5.2 The Likelihood Ratio Test

Compare $L(\hat{\theta})$ and $L(\tilde{\theta})$. If $h(\theta_0) = 0$ then $L(\hat{\theta})$ should be close to $L(\tilde{\theta})$. The test is base on the likelihood ratio $\lambda = \frac{\mathcal{L}(\tilde{\theta})}{\mathcal{L}(\hat{\theta})}$ with the test statistic

$$\mathcal{LR} = -2 \log(\lambda) = 2 \{ \log \mathcal{L}(\hat{\theta}) - \log \mathcal{L}(\tilde{\theta}) \} \quad (14)$$

We may encounter convergence problem when maximizing.

Under the null hypothesis,

$$\mathcal{LR} \xrightarrow{d} \chi^2(d) \quad (15)$$

1.5.3 The Langrage Multiplier or Score Test

Compare $S(\tilde{\theta})$ with 0 since $S(\hat{\theta}) = 0$. The test statistic is

$$\mathcal{LM} = S(\tilde{\theta})' [\mathcal{I}(\tilde{\theta})]^{-1} S(\tilde{\theta}) \frac{1}{n} \quad (16)$$

Under the null hypothesis,

$$\mathcal{LM} \xrightarrow{d} \chi^2(d) \quad (17)$$

The LM test works best in real life application. It is under weaker regulated conditions but it's still difficult to compute the $S(\theta)$ function.

第 2 章 Binary Choice Models

2.1 Linear probability model

In many applications, the variate of interest is binary, i.e., takes values 0 and 1. Let's denote $X = (X_1, \dots, X_k)'$ and the objective of the regression model is to estimate $E[Y|X]$ where $E[Y|X] = P(Y = 1|X)$.

In the **linear probability model** we assume that

$$P(Y = 1|X) = \beta_1 + \beta_2 X_2 + \dots + \beta_k X_k \quad (18)$$

thus the **interpretation of β_j** is the **change in the probability of success when x_j changes**.

$$\frac{\partial P(Y = 1|X)}{\partial X_j} = \beta_j, \quad j = 2, \dots, k \quad (19)$$

and the **predicted Y** is the **predicted probability of success**.

The linear probability model is estimated using **OLS**. But there is the potential problem that the fitted values can be outside $[0, 1]$. Even without predictions outside $[0, 1]$, we may estimate effects that imply a change in x changes the probability by more than +1 or -1. **$\hat{\beta}$ too big that would not make sense**. Also, this model **violates the assumption of homoskedasticity**, therefore we should use the Eicker-Huber-White **robust standard errors** to make inference. **Homoskedasticity states that $\text{Var}[Y|X]$ does not vary with x but here it necessarily depends on x .**

证明.

$$\text{Var}[Y|X] = \underbrace{E[Y^2|X]}_{(1)} - E[Y|X]^2$$

(1):

$$E[Y^2|X] = 1^2 \times P(Y = 1|X) + 0^2 \times P(Y = 0|X) = P(Y = 1|X)$$

Now we have,

$$\text{Var}[Y|X] = P(Y = 1|X) - P(Y = 1|X)^2 = \underbrace{P(Y = 1|X)}_{\beta_1 + \beta_2 x_2 + \dots + \beta_k x_k} [1 - P(Y = 1|X)]$$

So we see that the variance depends on the regressors. □

2.2 Index models for binary response